TrackerForAll for Unity

Introduction

TrackerForAll is a complete human tracking tool designed for Unity developers. It detects human pose, face, hand, and iris landmarks in real-time from webcam or video input using only an RGB camera — no additional hardware or depth sensors are required. It supports multiple platforms including Windows, macOS, Linux, Android, and iOS (with additional setup for Android/iOS cameras).

Based on detected landmarks, TrackerForAll can animate humanoid avatars to follow human movement, copy facial expressions, or replicate gestures. It also includes segmentation features to support advanced AR/VR use cases. The tool is configurable from the Unity Inspector and capable of running at up to 60 FPS.

Key Features

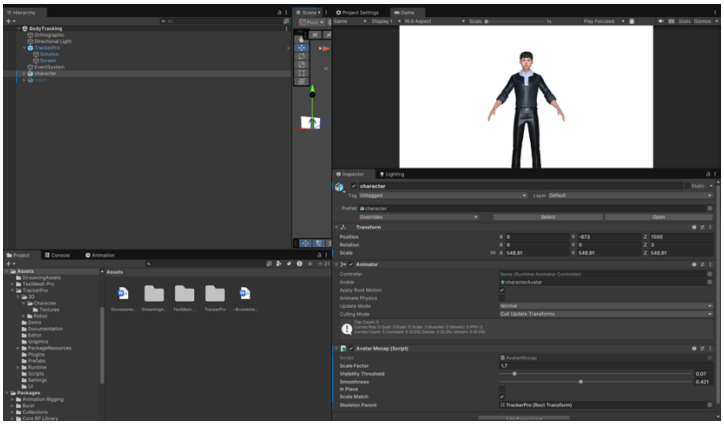
- **Pose Tracking**: Detects body landmarks in real-time to analyze posture and animate avatars. Outputs both 2D image and 3D world coordinates.
- **Face Tracking**: Detects faces and key features such as eyes, nose, and mouth in video streams.
- **Face Mesh Tracking**: Tracks 3D face landmarks and blendshape scores for facial expressions, virtual avatars, and effects.
- **Full Body Tracking**: Provides 543 landmarks combining body, face, and hands for holistic avatar motion.
- **Hand Tracking**: Detects hand landmarks, world coordinates, and handedness (left/right) for gesture-based control.
- **Iris Tracking**: Tracks eye iris movement in real-time, useful for virtual lens try-on and advanced interactions.
- **Hair Segmentation**: Segments hair in real-time video, useful for virtual coloring and styling.
- **Selfie Segmentation**: Separates users from their backgrounds for effects like virtual backgrounds, blurring, and immersive AR environments.
- **Avatar Support**: Works with any humanoid avatar (including ReadyPlayerMe). Supports auto-scaling, IK & FK constraints, and motion smoothness controls.
- **Cross-Platform**: Supports Windows, macOS, Linux, Android, and iOS (with camera configuration for mobile).

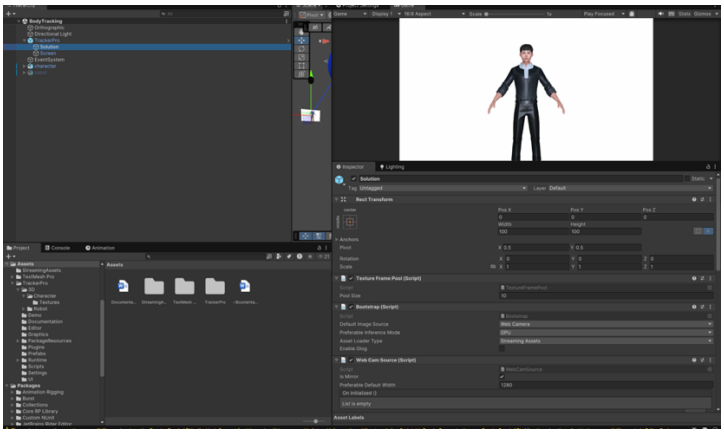
Use Cases

- Virtual Try-On (clothes, makeup, hair, contact lenses)
- Motion-Based Games
- Gesture-Based UI Interactions
- Humanoid Avatar Animation
- Face, Hand, and Pose Detection
- Avatar Motion Capture (Mocap)
- Creating Avatar Dance Videos from YouTube
- Fitness and Training Avatars
- AR/VR Virtual Presence Systems
- Live Streaming and Avatar Puppeteering
- Interactive Education and Training
- Remote Collaboration with Avatars

Project Setup

Step-by-Step Guide

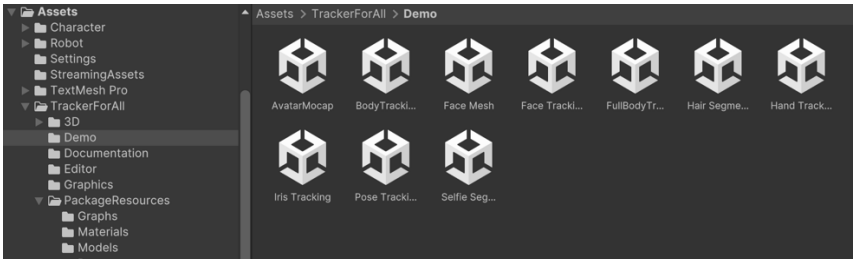




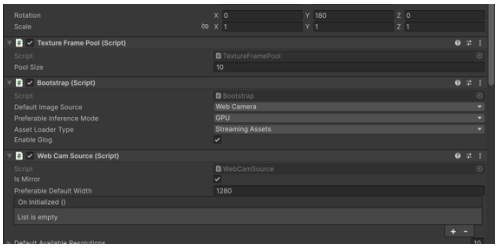
1. **Create a New Scene**
    - Open Unity and create a new scene.
2. **Add a Camera**
    - Add either an orthographic or perspective camera.
3. **Drag TrackerForAll Prefab**
    - Navigate to the `Prefabs` folder.
    - Drag `TrackerForAll` into your scene hierarchy.
4. **Import and Setup Avatar**
    - Import your humanoid avatar (e.g., ReadyPlayerMe).
    - Convert it into a humanoid rig in the Avatar import settings.
    - Add it into the scene.
5. **Configure TrackerForAll Scripts**
    - In `TrackerForAll > Solution` child, configure ML model scripts.
    - Define input source (video/webcam), choose model, and inference mode.
6. **Configure AvatarMocap Script**
    - Attach `AvatarMocap` script to your avatar.
    - Configure fields:

Field	Description
Scale Factor	Adjusts avatar's scale to match human
Visibility Threshold	Determines when a landmark starts tracking
Smoothness	Controls the movement interpolation
In Place	Whether the avatar changes position or not
Match Scale	Avatar resizes based on human-camera distance
Skeleton Parent	Parent transform of the landmark points

7. To run other Trackers check demo scenes



8. Choose backend type GPU if CPU do not work on android&iOS



Demo Scenes

- **PoseTrackingDemo** – Real-time body pose tracking and avatar animation
- **FaceTrackingDemo** – Face detection and facial key points
- **FaceMeshDemo** – 3D face mesh and expression tracking
- **FullBodyTrackingDemo** – Combines body, face, and hands for holistic motion
- **HandTrackingDemo** – Gesture-based control and hand interaction
- **IrisTrackingDemo** – Real-time iris tracking and lens try-on
- **HairSegmentationDemo** – Virtual hair coloring and styling
- **SelfieSegmentationDemo** – Background removal, blurring, and virtual environments
- **AvatarMocap** – Avatar motion capture and mimic from video or webcam
- **BodyTracking** – Avatar pose replication from webcam input

Demo Scenes

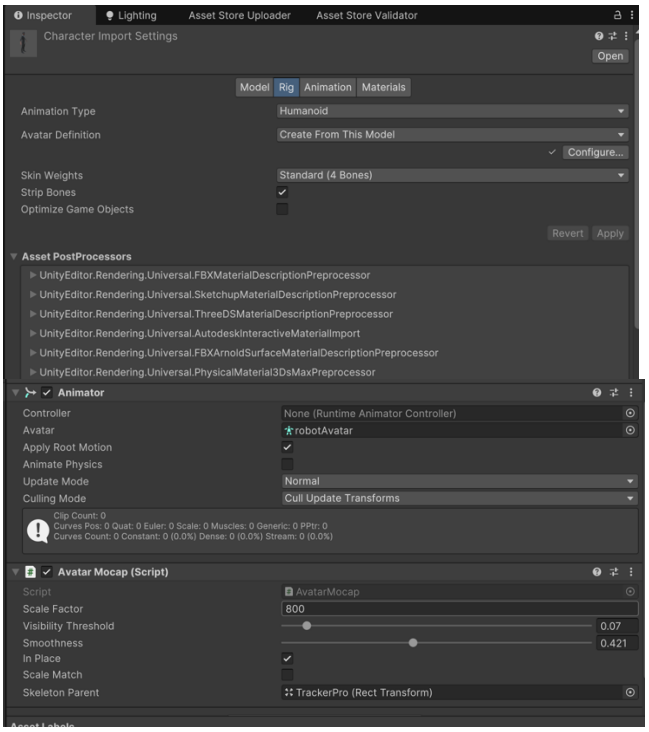
- **Assets Download**

  Download robot and character models,textures and prefabs from below link and import in your project. Convert the models into **humanoid** avatar.
  https://drive.google.com/file/d/1xpTdJwDmOL8ZilatmEcO_sRUbelo7x8o/view?usp=sharing

  https://drive.google.com/file/d/1DgtH_AT45bFw4gzE6JuEKSANO--_Ap0-/view?usp=sharing

  Video Source Link:
  https://drive.google.com/file/d/1Cx4qXr_O3l6I6EUU7H_hseDIr8ZDWnrK/view?usp=sharing

  Follow below screenshots to setup the avatar



Performance Tips

- Use the **Lite Model** for Android/iOS to improve inference speed.
- Use Low Resolution Input
- Ensure good lighting for accurate landmark detection.
- Adjust `Visibility Threshold` and `Smoothness` for natural avatar motion.

Support & Contact

📧 **Email**: realmoctopus@gmail.com
🌐 **Unity Asset Store Support Page**: Contact us via the TrackerForAll asset page

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Thank you for using **TrackerForAll** — the complete tracking toolkit for Unity. 🚀